Mary Palmer

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EDUCATION

Brigham Young University–Idaho Bachelor of Science in Economics Sep 2024 – Apr 2028
Minor: Data Science GPA: 4.00

- Full-Ride Merit Scholarship; Economics Society; Merchandising Society
- Teaching Assistant, Economics (Apr – Jul 2025): tutored students on microeconomics and coordinated weekly assignments for 65+ students
- Relevant Courses: Econometrics, Machine Learning, SQL, Business Analytics, Intermediate Statistics

EXPERIENCE

Senior Research Fellow – *BYU-Idaho, Rexburg, ID* Jan 2026 – Present

- Ran randomized experiments with 300+ participants, using fixed effects and clustered standard errors, leading the teacher to drop an unnecessary teaching practice
- Built 14 AI video tutorials now part of an official university course to support student learning
- Managed a team of research associates and oversaw survey deployment and treatment assignment

Data Science Intern – *Apruv, Remote* Jun 2026 – Sep 2026

- Wrote complex SQL queries on 10M+ rows of data to find intervention opportunities for a client
- Modeled intervention effects with fixed and random effects regression, identifying $76K/month in potential value

PROJECTS

Rural King – *Zone Pricing and Competitor Analysis, Consulting Project* Fall 2026 – Present

- Developing zone-pricing recommendations by clustering stores on demographics, competitor density, and climate, flagging under- and over-indexed markets
- Isolated a reported -18.5% price gap to -12.0% on exact-match products after finding two retailers drove most of it
- Built an interactive price-gap dashboard (HTML/CSS/JS, no framework) visualizing Rural King's competitive pricing position across 23 departments and 2,310 SKUs against 9 competitor retailers, with outlier-flagging logic to separate data noise from real pricing gaps
- Designed a four-signal scoring model to classify 164 stores into pricing-action tiers (raise/hold/sharpen) and flagged a threshold-design flaw before it was presented to stakeholders

The Econ Games 2026 – *National Analytics Competition, University of Kentucky, R* Winter 2026

- Collaborated with a 10-person BYU-Idaho economics team to analyze Equibase Co. data using economic reasoning and a fixed-effects logistic regression
- Built Power BI-style dashboards in R to identify performance trends, presented as a product pitch

Retail Promotion Pricing and Demand Analysis – *R / RStudio, AI-Simulated Data* Fall 2025

- Used two-way ANOVA and interaction testing on 3 years of AI-simulated weekly retail data to estimate the revenue impact of promotion depth
- Identified 25-35% promotion depth as a consistent revenue driver for a seasonal grocery product

Other Projects: Mutual Fund Analysis (Power BI), Religious Perceptions Survey (Power BI)

SKILLS & LEADERSHIP

Data Tools:

Python (sklearn, pandas, numpy), R, SQL (CTE, subqueries), GitHub, Power BI, Excel

Methods:

Statistical modeling, causal inference, hypothesis testing, data visualization

Leadership & Volunteer Work:

Economics Society: President; Distinguished Young Women: Mentor (12 participants), State First Runner-Up, $6,000+ in scholarships; Women's Church Group: President; Church Organist: 3 years